

DUALSense: An Open-Source Dual-User VR Platform for Driver–Worker Interaction with Multimodal Sensing

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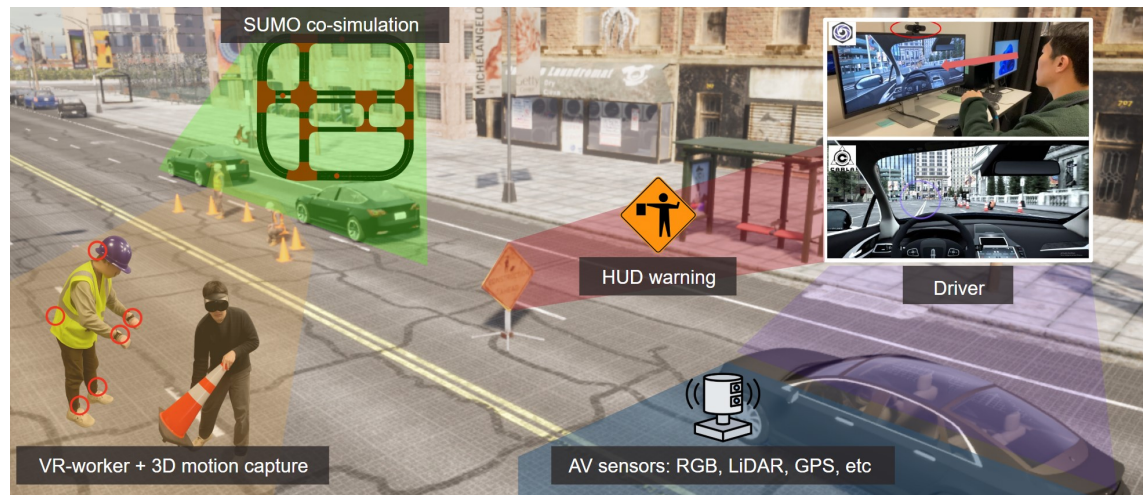


Fig. 1. Overview of our VR-based platform, highlighting driver–worker interaction, multimodal human-behavior capture, AV-sensor data collection, and co-simulation of vehicle and environment states.

Roadway work zone safety remains a critical challenge, as fatal crashes have continued to rise over the past decade. The behaviors of key road users—particularly drivers and on-site workers—are still poorly understood, largely due to the difficulty of capturing natural interactions in high-risk environments. To address this gap, we present DUALSense, an open-source virtual reality platform that supports dual-user interaction between a driver and a VR-based worker, enabling multimodal data collection in realistic construction-zone scenarios. The platform records worker motion and gaze, driver gaze behavior, as well as vehicle-centric data including trajectories, RGB images, LiDAR point clouds, and kinematic states. It further supports rapid and extensible scenario configuration, including traffic generation, weather and lighting variations, and worker behaviors derived from real human motion. A usability study (N=17) demonstrates that the platform supports naturalistic driving interactions with minimal discomfort, validating its suitability for roadway safety research and future autonomous-driving extensions.

Additional Key Words and Phrases: Virtual reality, driver–worker interaction, multimodal sensing, roadway zone safety, human factors

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1 Introduction

Roadway work zone safety has remained a critical challenge over the past decade. From 2014 to 2023, fatal work-zone crashes in the United States increased by 34%, a trend that persisted even during the pandemic. In 2023 alone, an estimated 101,000 work-zone crashes occurred [1], resulting in 899 deaths. A key issue is that current safety practices focus on retrospective inspections rather than proactive investigation of how drivers and workers actually behave in these environments. As national infrastructure continues to age, the demand for roadway construction is expected to increase, further elevating the urgency of enhancing work zone safety [2].

Implementing effective solutions for work zone safety requires overcoming several persistent challenges. First, driver attention—one of the primary contributors to safety risks—remains poorly understood, particularly during interactions with workers across diverse scenarios [3]. Second, real-world safety testing in active work zones is inherently hazardous and financially prohibitive [4, 5]. Third, existing virtual simulation tools fail to accurately represent dual-user interactions between drivers and workers. Fourth, as autonomous vehicles continue to develop, AVs will increasingly encounter work zones, yet high-quality multimodal data from these environments remain scarce [6]. To address these gaps, there is a clear need for a safe and controlled environment that can recreate dangerous roadway scenarios and capture detailed driver–worker interactions.

In this paper, we present DUALSense, an open-source VR platform for driver–worker interaction with multimodal sensing. The platform is built on top of CARLA [7] to maintain consistency with mainstream autonomous-driving simulation datasets. The following sections describe the system architecture, sensing components, and data collection methods. We then analyze drivers’ reactions across several high-risk work zone scenarios and compare behavioral differences between work zone types. On the VR-worker side, participants use a Meta Quest Pro headset to interact with the environment. Full-body motion is estimated through inverse kinematics, allowing workers to walk, gesture, and manipulate objects such as traffic cones using motion controllers. The system records worker trajectories, including position, rotation, and head orientation, which serves as a proxy for gaze direction. On the driver side, participants operate a vehicle using a Logitech gaming wheel paired with a custom driving interface. A self-developed webcam-based eye-tracking tool captures their gaze behavior. Although DUALSense supports AV takeover experiments [8], the present study focuses on naturalistic driving behavior. On the vehicle and environment side, the platform supports multiple sensor modalities, including four RGB cameras, a LiDAR sensor, vehicle trajectories, speed, and other tabular data. Semantic segmentation is directly obtained from CARLA to facilitate automated gaze–environment alignment. All sensor streams are synchronized by timestamp. Additionally, DUALSense incorporates an asynchronous co-simulation framework with SUMO [9] to control surrounding traffic and route behavior. Six different maps are used throughout the driving sessions and user studies to ensure scenario diversity. Figure 1 provides an overview of the proposed system and its integrated components.

The contributions of this research include

- (1) We developed **DUALSense**, the first open-source VR platform for roadway work zone safety that safely recreates hazardous work zone scenarios and supports synchronized multimodal data collection from both humans and AV-style sensors.

- (2) We designed a dual-user interaction framework that integrates full-body VR-worker motion, driver eye tracking, and vehicle perception, enabling realistic bidirectional driver-worker interactions that are not supported by existing simulation tools.
- (3) We conducted a user study with 17 participants in recreated high-risk roadway work zone scenarios to evaluate the usability and realism of the platform, and to demonstrate how the collected data can support analysis of drivers’ perceived difficulty and visual attention across different work zone types.

2 Related Works

In this section, we begin by reviewing applications of virtual reality in construction, including roadway work zones. We then discuss simulation platforms used in AV research, followed by driving simulators designed for studying human reactions and human factors outcomes. Finally, we compare existing dual- or multi-user VR platforms developed for transportation and AV research.

2.1 Virtual reality in construction and roadway work zones

Virtual reality has been widely adopted for safety research in construction and roadway work zones. By recreating safety-critical scenarios in an immersive yet risk-free environment, VR enables researchers to collect naturalistic reactions and behavioral responses from participants [10]. VR is often integrated with additional hardware—such as biometric sensors and exoskeleton systems—to capture workers’ stress levels, physiological states, and cognitive load [11, 12, 13]. It also provides a cost-efficient approach for testing work–machine and worker–robot interactions [14, 15, 16]. For roadway work zone safety, VR and augmented reality (AR) serve as important tools for creating interactive roadside simulations in which workers can grab cones, manipulate tools, and guide traffic in the presence of surrounding vehicles. [17, 18, 19, 20]. Several systems also incorporate and evaluate novel warning or alerting mechanisms designed to protect workers from approaching vehicles [21, 22]. These platforms enable the collection of rich multimodal data—including reaction times, behaviors, and movement trajectories—from participants. Such data supports learning-based methods for validating warning-system designs and predicting workers’ responses to potential hazards.

2.2 Simulation platforms for autonomous vehicle research

Due to the high cost of testing AV algorithms in the real world, researchers heavily rely on simulation-based datasets and often validate their proposed methods in simulation platforms [7]. Depending on different settings and research goals, many simulators have been developed. Vehicle-to-Vehicle (V2V) and Vehicle-to-Everything (V2X) applications require data from multiple agents, especially perception data. To support these needs, frameworks such as OpenCDA, OpenV2V, and V2X-Sim are built on top of CARLA, providing datasets, simulations, and reference code [23, 24, 25]. MetaDrive [26] addresses the lack of simulators designed for reinforcement learning by enabling high-frequency interaction with the environment. Other simulators offer scalable capabilities for automated asset generation in urban navigation and micromobility scenarios [27, 28].

Beyond closed-loop simulators with physics engines, generative world models and open-loop systems—such as the Gaia series and Nvidia Omniverse—provide new ways to simulate complex environments [29, 30]. To our knowledge, ROADWork [31] is the only publicly available dataset targeting roadway work zones. Built on top of the Michelin Mobility Intelligence (MMI) Open Dataset [32], ROADWork provides RGB images, weakly aligned GPS data, and language annotations for scene understanding. However, it lacks ground-truth trajectories; the trajectories it offers

are reconstructed from images. Therefore, there remains a significant gap in datasets for AV research in work zone environments—specifically those that include RGB images, ground-truth trajectories, and vehicle control signals.

2.3 Driving simulators for data collection and human factors

Driving simulators are widely used for collecting data from participants in data-driven transportation research and human factors studies. Remonda et al. [33] developed a racing simulator to gather human driving data for improving autonomous racing agents. Human drivers can also provide real-time interventions that serve as rewards within reinforcement learning frameworks [34, 35]. Under safety-critical scenarios, simulators allow researchers to safely collect vehicle trajectory data from human drivers for risk-aware AV modeling [36, 37].

Driving simulators are frequently used to evaluate driver state, cognitive load, and human-machine interaction systems due to their safety, scalability, and cost-efficiency. To capture human responses, sensors such as eye trackers, electrodermal activity (EDA), electroencephalography (EEG), and other physiological monitors are commonly integrated. For work zone safety, Ye et al. [38] used a VR driving simulator to evaluate in-vehicle HUD warning systems on freeways. ARive [39, 40] combined an AR headset with a driving simulator to study risk-zone alerts and their effects on distraction and fatigue. To evaluate new visualization techniques for car-following behavior, Sheng et al. [41] conducted user studies using a driving simulator. Additionally, Costa et al. [42] demonstrated the effectiveness of a conversational AI racing coach using a CARLA-based simulation environment. In contrast, TimelyTale [43] relies on real-world data collection to capture human user reactions using onboard AV sensors. While this approach enables high-fidelity observations, it incurs substantially higher data collection costs compared to VR-based simulation, particularly for safety-critical scenarios.

2.4 Multi-user VR platforms for transportation and AV studies

In complex urban environments, multi-user simulation is essential for capturing realistic interactions between diverse road users and for collecting data that can faithfully recreate dynamic scenarios. Prior VR-based systems have incorporated various road users—including pedestrians, cyclists, e-scooter riders, and mixed traffic participants—to study interactions with human drivers or autonomous vehicles.

For pedestrian-vehicle interaction, Crosato et al. [44, 45] introduced a platform where a VR pedestrian, tracked with a Perception Neuron motion-capture system, interacts with a driver using a gaming wheel in an Unreal Engine environment. Their system supports trajectory recording for both the pedestrian and the human-controlled vehicle. JaywalkerVR [46, 47] similarly provides an AV-pedestrian interaction platform built on CARLA, offering synchronized trajectory data from both agents.

For cyclist-vehicle interaction, Angulo et al. [48] developed a VR cycling platform to evaluate roadway design alternatives using cyclist gaze behavior, biometric responses, body pose, and trajectory data. For e-scooters, DualTake [49] proposed a dual-user takeover platform that enables interaction between ADAS takeover behaviors and e-scooter riders, although the system is based on 360° video captured using AirSim [50] rather than full physics-based simulation.

For multi-pedestrian scenarios, Feng et al. [51] presented a dual-pedestrian VR platform to study AV external HMI designs. For mixed-user environments, StrangeLand [52, 53, 54, 55] supports interactions among multiple drivers and pedestrians and is publicly available and implemented in Unity. Kang et al. [56, 57] further proposed an open-source multi-user VR platform enabling interactions among drivers, pedestrians, and cyclists, with support for trajectory logging, biometric sensing, and AV eHMI evaluation.

Platform / Study	Supported Users	Road	Collected Data	Open Source	Software	Extendable to Work Zones
Crosato et al. [44, 45]	Driver + Pedestrian		Trajectories (pedestrian & vehicle)	No	Unreal	<i>Potential</i>
JaywalkerVR [46, 47]	AV + Pedestrian		Trajectories (pedestrian & vehicle)	No	CARLA (Unreal)	<i>Limited</i>
Angulo et al. [48]	Cyclist + Vehicle		Gaze, biometrics, pose, trajectories	No	Unity	<i>Limited</i>
DualTake [49]	ADAS Driver + E-Scooter		Takeover behaviors	No	AirSim (Unreal)	<i>Limited</i>
Feng et al. [51]	2 Pedestrians + AV		Pedestrian trajectories, gaze cues	No	Unreal	<i>Limited</i>
StrangeLand [52]	Multiple Drivers + Pedestrians		Driver trajectories, pedestrian actions	Yes	Unity	<i>Potential</i>
Kang et al. [56]	Driver + Pedestrian + Cyclist		Trajectories, biometrics, eHMI interaction logs	Yes	Unity	<i>Potential</i>

Table 1. Summary of multi-user VR platforms for transportation and AV studies.

Table 1 summarizes existing dual- and multi-user platforms used in transportation, urban safety, and autonomous-vehicle research. None of the publicly available systems support construction-site safety research, particularly scenarios involving workers, who exhibit far more degrees of freedom than other road users such as pedestrians. In addition, most platforms emphasize trajectory data rather than human-reaction measurements (e.g., gaze, heart rate, EDA), and none capture vehicle-centric sensor data such as RGB images or point clouds. Considering the widespread use of simulation in AV testing and data collection, the prevalence of driving simulators in human-factors research, and the growing adoption of VR for construction and worker safety studies, this gap becomes even more apparent. Overall, the field lacks a VR-based simulation platform that is purpose-built for construction-site research while also supporting AV research through self-driving-compatible multimodal datasets.

3 Methodology

This section introduces DUALSense, an open-source dual-user VR platform for driver–worker interaction with multimodal sensing. DUALSense supports real-time interaction between the driver and the VR-based worker, along with synchronized multimodal data collection including trajectories, gaze movement, biometric signals, RGB images, LiDAR point clouds, and semantic segmentation outputs. The platform also enables rapid scenario configuration, SUMO-based co-simulation for generating realistic traffic, and 3D motion-capture-driven animation. DUALSense is designed for human-factors research, transportation safety studies, and autonomous driving data collection. The system is built on top of CARLA 0.9.15, a widely used version that ensures compatibility with existing CARLA-based benchmarks and supports future extensions.

Figure 2 illustrates the overall system pipeline, consisting of four key components: (1) VR-worker module: full-body inverse kinematics, motion-driven animation, and cone manipulation using VR controllers, with gaze and trajectory logging. (2) Driver interface: a driving UI that integrates a gaming steering wheel and a low-cost webcam-based eye-tracking system. (3) Sensor data collection: four RGB cameras, one LiDAR, vehicle-state data (trajectory, yaw angle, steering, throttle, brake), and driver-related data including UI screen capture and semantic segmentation images aligned with gaze. (4) Asynchronous CARLA–SUMO co-simulation: supporting realistic traffic generation, route planning, and

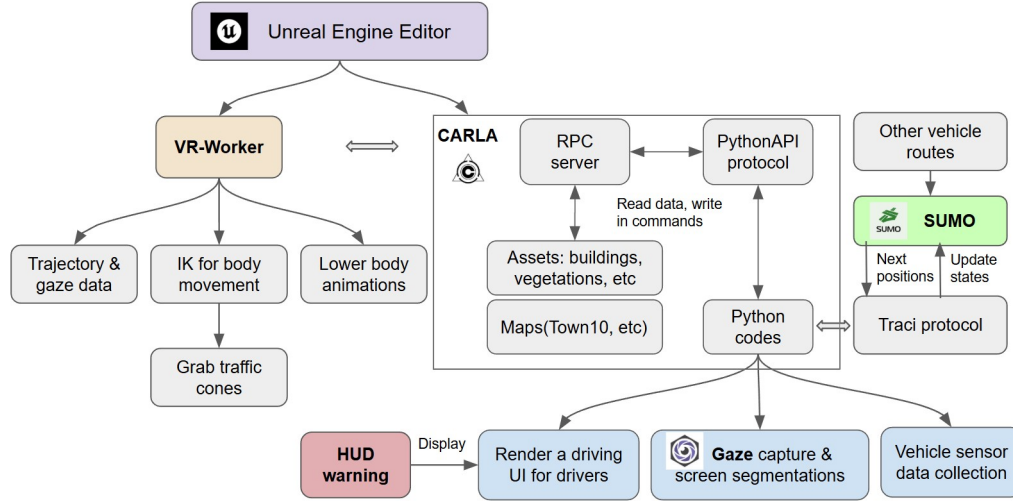


Fig. 2. Overview of the DRIVE platform. (a) VR worker with trajectory capture, full-body animation, and object grabbing; (b) AV-style sensor data collection; (c) driving interface and gaze capture for drivers; and (d) CARLA–SUMO co-simulation.

custom work-zone map design, suitable for driving-simulator–level experiments. All components are open source and will be described in detail in the following subsections.

3.1 VR-worker module

VR headset can provide an immersive environment for users, which is fit for roadusers especially necessary for those who need to walk on their own feet. For roadway workers, they are supposed to walk by their own feet, also grabbing traffic cones and tools with their hands. To fit for that, there are five parts: (a) full-body movement calculation, (b) lower body animations, (c) grabbing with motion controllers, (d) pre-recorded various worker animations, (e) trajectory and gaze data capture.

Full-body IK and lower-body animation logic follow Voeten’s tutorial [58], and locomotion poses are from the Advanced Locomotion System V4 [59]. We use the Meta Quest Pro; its IMU-based position and rotation streams drive IK and motion updates. Two thresholds (movement and rotation) determine whether the avatar should update its transform. A desired player height is defined, and interpolation smooths position and rotation changes to reduce jitter from the driver’s perspective.

Upper and lower body behaviors are handled separately. The upper body follows the headset and controllers, and Unreal’s IK system computes arm motion automatically. The lower body uses 25 custom walk/run poses organized in a Blueprint-based BlendSpace. After smoothing speed and position, the system updates locomotion using the user’s head-tracked speed, position change, and rotation, producing stable and realistic leg motion.

Grabbing is triggered when a motion controller approaches a grabbable object (e.g., traffic cones). Pressing the grab button attaches the object; releasing it lets physics drop the item naturally. To support one or multiple workers with varied behaviors, we also captured full-body animations using Sony Mocopi and Blender [60]. Appendix A provides example screenshots illustrating steps (a), (b), (c), and (d).

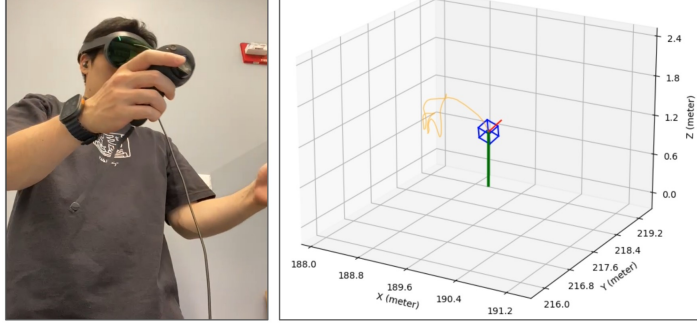


Fig. 3. VR worker data capture: left shows the user in the immersive VR environment, and right shows the visualization of the recorded 3D trajectory and gaze direction.

The VR movement and locomotion system is implemented entirely using Unreal Engine Blueprints. However, to enable efficient data capture from VR users, we implemented additional C++ modules for higher-performance input and output handling. We record the full transform of each VR worker and decompose it into XYZ location and rotation. Because workers often bend, bow, or squat, the Z-axis height information is essential; unlike prior pedestrian-vehicle interaction studies that typically reduce trajectories to 2D XY coordinates, our setting requires full 3D tracking. Gaze movement is also critical for characterizing worker attention and providing additional cues for trajectory prediction. Directly using motion-controller orientation introduces substantial noise [22], so we represent gaze direction using the headset orientation, following prior work [22, 61]. The data collection frequency is set to 5 Hz, which is sufficient for road users walking on foot, as their movement speed is significantly lower than that of vehicles. Figure 3 illustrates the VR user (left) and the visualization of XYZ trajectory and gaze direction (right).

3.2 Driver interface

Work zone crashes and near-miss events are often caused by driver- or vehicle-related factors; therefore, our data collection pipeline focuses more heavily on the driver side. The human-centered data include gaze movement, heart rate, EDA, and control signals captured from the steering wheel and pedals. To ensure a realistic driving experience, the driving UI is optimized to maintain a minimum frame rate of 35 FPS, which is sufficient for human users. Drivers operate the simulation using a monitor and a Logitech gaming wheel, as shown in Figure 4(a). The UI design includes a rear-view mirror at the top center of the screen and a speedometer (mph) at the bottom center. A webcam-based eye tracker is mounted on the top of the monitor, illustrated in Figure 4(b). To synchronize gaze with the driving interface, the system records UI screenshots at 10 Hz in low resolution to reduce rendering overhead. Because the overall system is human-centered, both the driver and the VR worker must perceive the environment as realistic. CARLA’s native semantic segmentation further enables abstracting gaze points into 2D screen-space coordinates, allowing automated identification of the object classes that drivers attend to—an important feature for both human-factors analysis and AV imitation-learning research. Examples are shown in Figure 4(c) and (d).

Gaze behavior is critical for understanding driver attention and situational awareness [62, 63]. However, most gaze-tracking systems used in driving simulators are expensive [64, 65, 66]. To address this issue, we propose a cost-efficient, open-source, webcam-based eye-tracking tool for screen-based driving simulators. Due to inter-individual differences in eye characteristics, a commercial calibration tool is still needed to ensure accuracy. We use the Beam Eye Tracker

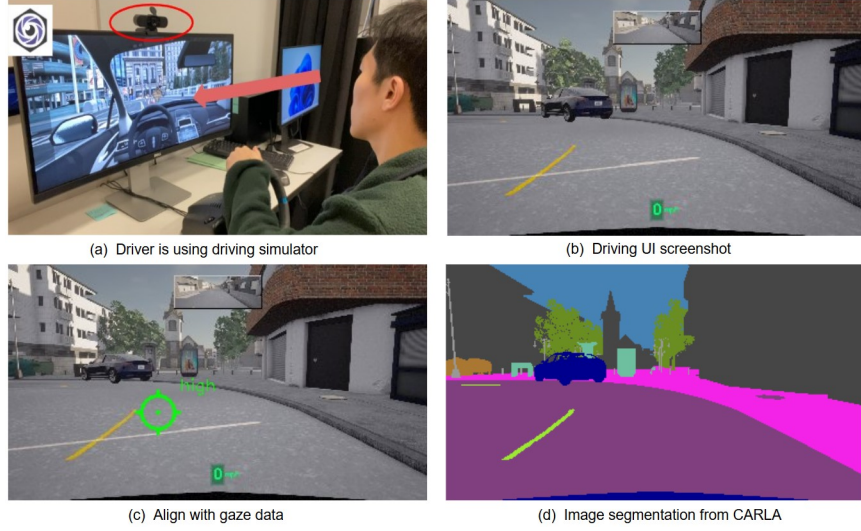


Fig. 4. Driving UI and gaze capture workflow: (a) driver interacting with the system, (b) driving UI design, (c) gaze data aligned with UI screenshots via timestamps, and (d) semantic segmentation outputs from CARLA.

for calibration, and with its SDK, we developed a 10 Hz gaze-trajectory logger with confidence scores. The logging frequency aligns with the 10 Hz UI screenshots, and gaze capture is implemented in a multi-threaded manner with adjustable frequency.

Finally, vehicle control is essential for providing realistic haptic feedback and maintaining driver immersion. The control signals from the Logitech wheel and pedals are normalized, with steering mapped to $[-1, 1]$ and throttle/brake mapped to $[0, 1]$, ensuring compatibility with motion-planning-based AV research. The vehicle-control module is implemented using CARLA’s official control API. In addition, we integrate CARLA’s Level-4 autopilot as an optional autonomous-driving mode, which can be extended for future AV takeover or human-automation interaction studies. However, since our focus is on naturalistic human driving in work-zone scenarios, only human-driven trials are used in the present user study.

3.3 Sensor data collection

In addition to the sensing modalities described earlier, the system adopts a multi-threaded asynchronous architecture to ensure stable and high-throughput multimodal data collection during driving simulation. Sensor acquisition is fully decoupled from disk I/O operations: each modality (e.g., RGB cameras, LiDAR, UI recordings, gaze tracking, and semantic labels) is captured in its own lightweight thread, where incoming data are first placed into thread-safe queues before being written to disk by background workers. This design prevents temporary saving delays from impacting real-time simulation performance.

All data streams use a unified timestamp and shared frame indices to support temporal alignment across modalities. To maintain responsiveness under variable system load, the queues employ a back-pressure strategy in which the oldest entries are discarded when approaching capacity, prioritizing real-time operation over lossless logging. Several optimizations are incorporated to reduce overhead, including downsampling UI recordings, staggering sensor sampling frequencies, and isolating gaze logging from rendering tasks. With these measures, the system achieves stable multimodal

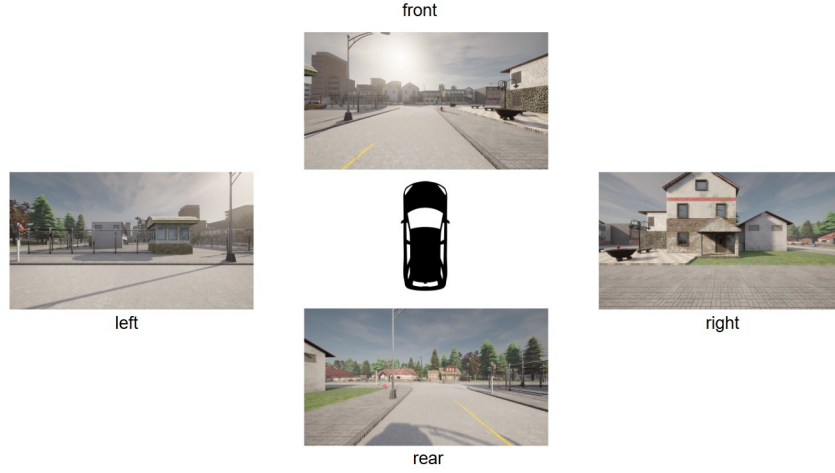


Fig. 5. Sensor data collection: four RGB cameras demo

throughput while maintaining a consistent 35 FPS rendering rate in CARLA’s asynchronous mode, ensuring that data collection does not interfere with the interactive driving experience. The detailed architecture is shown in Appendix B.

Moreover, in AV research, the RGB images play the most important role in visual perception. In the previous simulation based AV benchmark dataset, especially who use the bird-eye-view (BEV) reconstruction, they applied four to six cameras to cover the 360 degrees of vehicle surroundings [67, 68, 69]. Therefore, we design four camera surrounding the vehicle in 10 Hz shown in Figure 5. LiDAR can generate point cloud data, providing depth information, plays an important role in the perception system. we create one LiDAR on the top of vehicle to scan 360 degrees, 5 Hz. Besides that, we also collect some tabular data in 10 Hz, including XYZ trajectory, yaw, speed, accelerations. The other sensor demo visualization images are shown in Appendix B.

3.4 Asynchronous CARLA-SUMO co-simulation

For co-simulation, DUALSense supports realistic traffic generation, route planning, and custom work-zone map design, making it suitable for driving-simulator experiments. We integrate SUMO (Simulation of Urban MObility), an open-source microscopic traffic simulator widely used for modeling large-scale vehicle flows, which provides more reliable and configurable traffic behavior compared to CARLA’s native traffic manager.

In the official CARLA-SUMO co-simulation framework, both systems run in synchronous mode. However, synchronous stepping is incompatible with our VR-based driver simulator, where high-frequency rendering (35 FPS) and responsive UI updates are required. Moreover, in our setup, SUMO controls all non-ego vehicles while CARLA handles 3D visualization and physics. The ego vehicle, driven by a human user, must remain decoupled from SUMO control while still interacting with traffic simulated in SUMO. To meet these requirements, we develop an asynchronous two-way CARLA-SUMO co-simulation scheme.

In this architecture, SUMO updates the positions and headings of surrounding vehicles and sends them to CARLA at 20 Hz. CARLA then updates these vehicles’ transforms while disabling physics to avoid conflicts. Conversely, CARLA sends the ego vehicle’s position back to SUMO so that SUMO vehicles can perceive the ego car and react to avoid

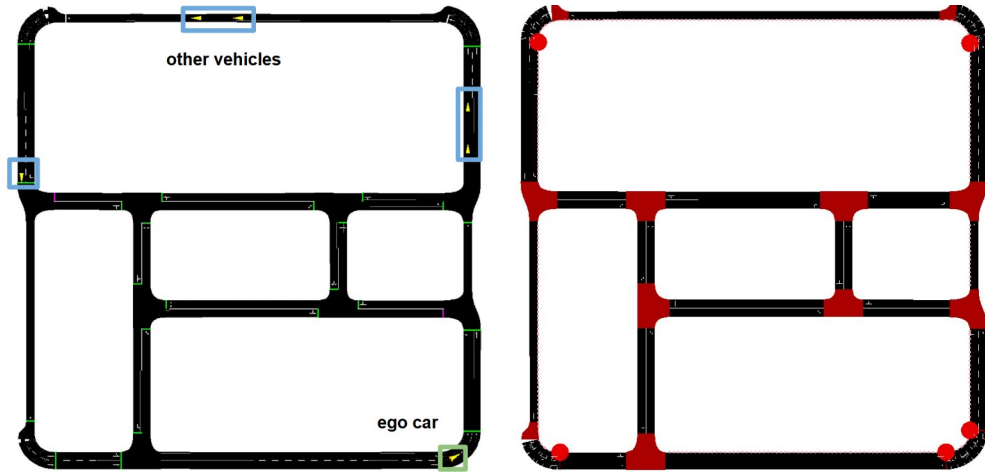


Fig. 6. CARLA-SUMO co-simulation: the SUMO GUI for traffic visualization (left) and NetEdit for network and work-zone lane configuration (right).

collisions. This bidirectional exchange enables realistic traffic interactions while allowing the human-driven ego vehicle to operate freely, the SUMO GUI is shown as Figure 6 left.

For roadway work zones, static work-zone layouts are placed in the CARLA 3D scene. To ensure SUMO vehicles are aware of work-zone constraints, corresponding lanes in the SUMO network are closed using NetEdit (Figure 6 right), enabling smooth rerouting around work zones. Work zones in CARLA include warning signs, traffic cones, barrels, and both pre-recorded and VR-driven worker animations, allowing naturalistic representation of short-term and temporary work zones—conditions known to be high-risk and lacking standardized layouts.

Scenario design is essential to representing diverse and hazardous conditions. Based on crash-frequency research [70, 71], we implement five dangerous work-zone scenarios: nighttime, rainy conditions, poor signage, roadway curvature, and truck-involved zones. Nighttime and rain are created by modifying CARLA’s lighting and weather parameters. Poor-signage scenarios remove advance warning signs, which commonly occurs on narrow urban streets where signage placement is constrained. Curvature scenarios place work zones on road bends, and truck-involved scenarios replace surrounding cars with truck models. To avoid route memorization effects across participants, we use multiple CARLA maps (Town 02, Town 04, Town 05, and Town 10HD), reusing some maps with different trajectories. Additional traffic signs such as “go straight,” “left turn,” and “right turn” are placed to guide drivers. Figure 7 illustrates the six scenario configurations.

Finally, to ensure stable real-time performance, the entire platform is packaged as a standalone executable from the Unreal Engine editor, enabling consistent, high-performance operation during user studies.

4 Usability Testing Results

The proposed platform is designed to support roadway work zone safety research by enabling naturalistic human–vehicle–environment interactions in controlled yet realistic settings. The primary goal of the user study is to validate whether the platform can be reliably used by human participants and whether it is capable of revealing meaningful behavioral differences relevant to work-zone safety research.



Fig. 7. CARLA-SUMO co-simulation: various recreated work-zone scenarios based on crash-frequency analyses.

To investigate how hazardous work-zone conditions influence driver responses, we compare drivers' behavior across multiple challenging scenarios and benchmark them against a normal work-zone baseline. These scenarios introduce risk factors commonly observed in real-world work zones, such as reduced visibility, complex road geometry, and the presence of heavy vehicles. In addition, we examine whether different work-zone designs and interaction modes lead to measurable changes in drivers' attention and driving behavior.

Prior work has primarily focused on driver interactions with static warning signs or pre-recorded worker animations. In contrast, our platform introduces interactive VR-based workers, enabling bidirectional interactions between drivers and workers. This allows us to examine whether more realistic and dynamic work-zone elements elicit different driver responses compared to conventional designs. Since improving work-zone safety often depends on drivers' timely perception and reaction to hazards, we also consider the role of vehicle-to-infrastructure (V2I) warnings as an additional design factor that may influence driver behavior.

Our main hypothesis is that the proposed DUALSense platform can capture scenario- and design-dependent differences in drivers' attention and driving behavior through synchronized multimodal sensing, including gaze and vehicle kinematics. Specifically, we expect that both hazardous scenario characteristics and work-zone design choices will affect drivers' visual attention patterns and longitudinal driving behavior while passing through the work zone. Accordingly, this usability study addresses the following research questions:

- **RQ1:** Do hazardous work-zone scenarios influence drivers' attention and driving behavior?
Null hypothesis ($H_{0,1}$): Hazardous scenarios have no significant effect on drivers' attention (e.g., gaze duration and fixation ratio) or driving behavior (e.g., weighted mean and variability of speed and acceleration).
- **RQ2:** Do different work-zone designs and interaction modes alter drivers' attention and longitudinal driving behavior?
Null hypothesis ($H_{0,2}$): Work-zone design and interaction mode have no significant effect on drivers' attention or longitudinal driving behavior.

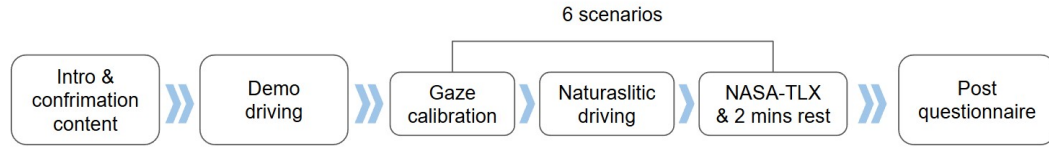


Fig. 8. Pipeline of the user study

4.1 User study design

Participant recruitment was conducted under an approved IRB protocol. All participants were required to hold a valid driver's license. A total of 17 drivers (13 male, 4 female) were recruited; data from one participant were excluded due to sensor instability, resulting in 16 participants included in the analysis. The system always involved two people in real time: a driver and a VR worker. All participants served as drivers, while the VR worker role was performed by one of two civil engineering students familiar with roadway work-zone procedures.

The study pipeline is illustrated in Figure 8. First, participants were briefed on the driving tasks and completed the consent form. Participants then completed a short demo driving scenario to become familiar with the simulator; data from this demo were not used in the analysis. Afterward, participants rested for at least 2 minutes before proceeding to the formal study.

Each participant completed six driving scenarios: one normal baseline scenario and five dangerous scenarios (Figure 7). The normal scenario always appeared first and included good lighting, no rain, appropriate warning signage, and surrounding vehicles (Tesla Model 3) on straight road segments. The five dangerous scenarios were designed such that only one hazardous factor varied at a time. For each scenario, the procedure remained consistent: (1) perform gaze calibration using the Beam eye-tracking system; (2) adjust the steering wheel tightness for comfortable control; (3) complete the driving task; (4) fill out a NASA-TLX form to evaluate cognitive load; and (5) rest for at least 2 minutes before proceeding to the next scenario. After completing all six scenarios, participants filled out a final post-study questionnaire.

For scenario design, each driving scenario uses a distinct map layout, and traffic lights alternate between green and red every 20 seconds. The number of surrounding vehicles ranges from four to five depending on the map size. All surrounding vehicles are Tesla Model 3 cars, except in the truck-involved scenario, where CARLA trucks are used. Six scenarios were designed for data collection: a normal work zone (S1), a work zone without warning signage (S2), a rainy condition (S3), a work zone located on a curved road rather than a straight segment (S4), a nighttime condition (S5), and a truck-involved scenario in which all Tesla Model 3 vehicles are replaced with trucks (S6).

Each scenario contains three work zones, assigned at separate locations. The work-zone designs are as follows: (1) Normal work zone: two prerecorded animated workers operate safely within the work-zone boundary (one squatting for pavement work and one standing and idling). (2) VR-worker work zone: the VR worker replaces traffic cones to create a temporary work-zone layout in real time. (3) Normal work zone with V2I: identical to the first work zone but includes a V2I warning displayed to the driver when they are 100 meters upstream. For each work zone, the active working area is 8 meters long, with a stationary warning sign placed 8 meters before the first traffic cone—except in the poor-signage scenario, where no stationary warning signage is provided. This design reflects real narrow urban conditions where signage may be missing, such as on New York City streets. Figure 9 illustrates the visual differences among the three work-zone designs.

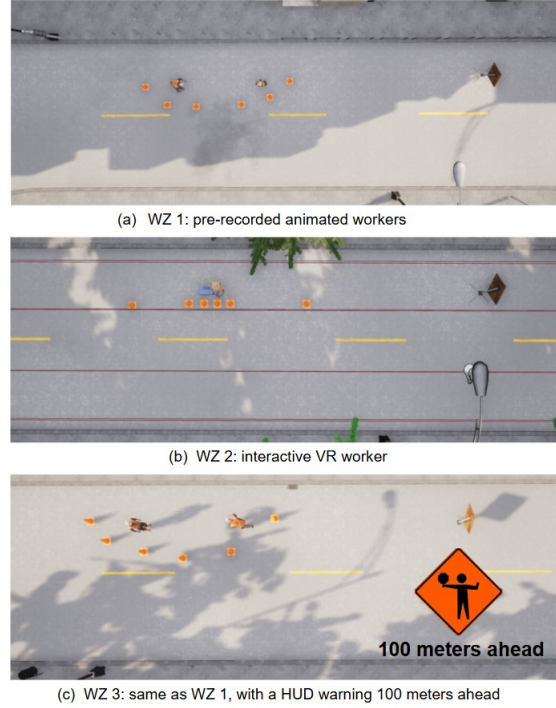


Fig. 9. Work-zone configurations used in the usability study. (a) WZ1: baseline work zone with traffic cones and pre-recorded animated workers; (b) WZ2: work zone with an interactive VR worker; (c) WZ3: same as WZ1, with an additional HUD warning displayed 100 meters ahead.

4.2 Metrics & Data Preparation

Gaze information has played a primary role in characterizing drivers' perception in prior research [3, 60]. Among these metrics, gaze duration required manual annotation, while all other gaze and vehicle-based metrics were automatically derived from synchronized raw sensor data using timestamps. To assess drivers' perceptual responses in roadway work zones, we employed two eye-tracking metrics: gaze duration and gaze fixation ratio. Gaze duration is defined as the time interval between a driver's first fixation on the target work zone (t_f) and the moment the vehicle passes through the work zone (t_p), as formulated in Equation 1.

The gaze fixation ratio measures the proportion of fixations directed toward specific work-zone elements—such as warning signs, traffic cones, and workers—relative to the total number of fixations within the work zone. As shown in Equation 2, n_f denotes the number of fixations on target work-zone elements, and N represents the total number of fixations recorded within the work zone. To ensure reliable annotation, screenshots were automatically captured at a frequency of 10 frames per second.

To capture spatial variations in drivers' visual attention, gaze fixation ratios were computed using three concentric regions centered on the work zone in image space, with radii of 50, 100, and 120 pixels (denoted as R50, R100, and R120). Smaller radii reflect attention allocation to immediate hazards, while larger radii capture anticipatory and peripheral

visual monitoring as drivers approach and traverse the work zone. Using multiple spatial scales allows us to examine how attention distribution changes with scenario hazard level and work-zone design.

For driving behavior, we focus on vehicle kinematics, using speed and acceleration to characterize drivers' responses to work zones. For each work-zone traversal, speed and acceleration signals were temporally segmented based on the duration of the work zone and aggregated into summary statistics, including the mean (speed mean, acceleration mean) and variability (speed standard deviation, acceleration standard deviation).

To assess cognitive load, we collected self-reported measures targeting both scenario-level and work-zone-level effects. After completing each driving scenario, participants rested for two minutes and then completed a NASA-TLX questionnaire to report their perceived cognitive workload during that scenario. After all driving tasks were completed, participants filled out a post-study questionnaire. One section of this questionnaire evaluated different work-zone designs to avoid priming effects or carryover bias during subsequent driving tasks. This section included three questions assessing perceived clarity, predictability, and attention required for each work-zone type. Additional details of the questionnaires are provided in Appendix C.

$$\text{Gaze Duration} = t_p - t_f \quad (1)$$

$$\text{Fixation Ratio} = \frac{n_f}{N} \quad (2)$$

4.3 Scenario-Level Effects on Driver Attention and Behavior

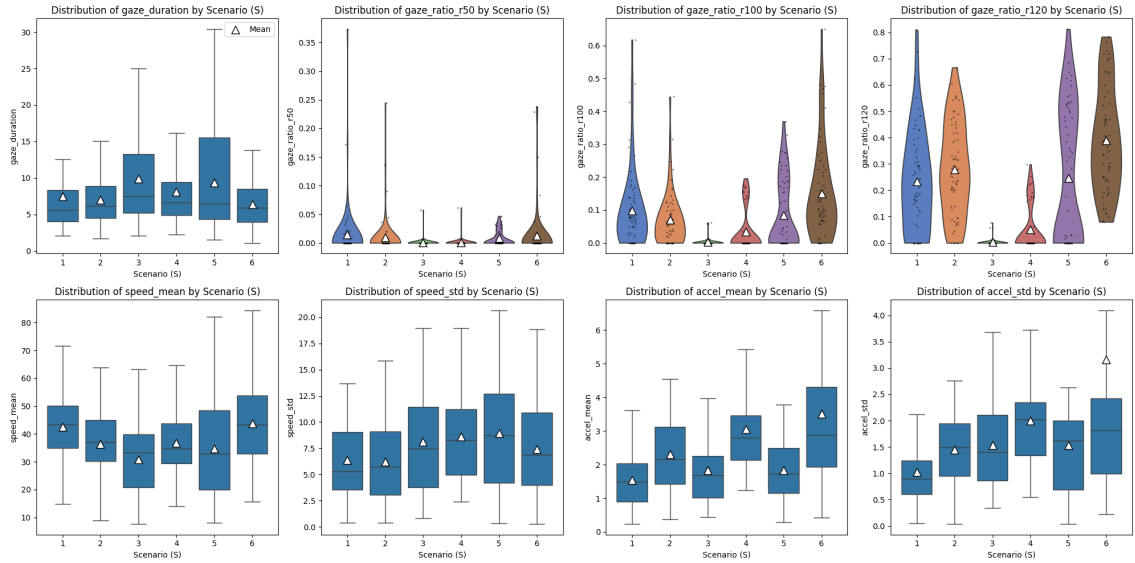


Fig. 10. Scenario-level distributions of driver attention and longitudinal driving behavior. (a)–(d) gaze-based attention metrics; (e)–(h) speed and acceleration metrics across six scenarios (S1–S6). Triangles denote mean values.

To examine whether different hazardous scenarios lead to distinct driver responses, we conducted a one-way ANOVA across scenarios for all objective and subjective metrics. A one-way ANOVA is suitable here because each metric is

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compared across multiple independent scenario conditions with a single categorical factor (scenario type). Table 2 summarizes the ANOVA results.

4.3.1 Attention Metrics. As shown in Fig. 10(a), gaze duration varies across scenarios. Compared to the baseline work zone (S1), drivers show longer gaze duration in the rainy scenario (S3) and the nighttime scenario (S5), indicating increased visual processing demands under degraded visibility. Within-scenario variability further suggests individual differences in visual adaptation strategies.

Figures 10(b–d) show gaze fixation ratios computed at three spatial scales (R50, R100, and R120). At R50, fixation ratios remain close to zero across all scenarios, with only marginal differences, as drivers rarely fixate directly on workers or signage during continuous driving.

At larger radii (R100 and R120), fixation ratios differ clearly across scenarios. In the rainy scenario (S3), lower fixation ratios are likely caused by reduced visibility, which limits reliable visual engagement with work-zone elements. In the curved-road scenario (S4), drivers tend to allocate more attention to lane markings and road geometry during turning maneuvers, leading to lower fixation ratios on work-zone elements. In contrast, the baseline scenario (S1), the no-warning-signage scenario (S2), and the nighttime scenario (S5) exhibit higher fixation ratios. The truck-involved scenario (S6) shows the highest fixation ratios, indicating substantially increased visual demand.

4.3.2 Driving Behavior Metrics. As shown in Fig. 10(e–f), the baseline scenario (S1) has the highest mean speed and the lowest speed variability, reflecting stable and confident driving. The nighttime scenario (S5) shows lower mean speed but higher variability, consistent with cautious yet less stable speed control. The rainy scenario (S3) also shows reduced speed, highlighting the impact of weather conditions.

Figures 10(g–h) show acceleration metrics across scenarios. The baseline scenario (S1) exhibits both low acceleration mean and variability. The truck-involved scenario (S6) shows the highest acceleration mean and variability, which aligns with observed overtaking attempts and abrupt longitudinal control. Elevated acceleration is also observed in the curved-road scenario (S4) and the no-warning-signage scenario (S2), likely due to geometric constraints and reduced anticipatory information.

4.3.3 Self-Reported Cognitive Load. NASA-TLX scores range from 1 (very low) to 5 (very high). Among all subjective metrics, only *Effort* shows marginal significance ($F = 2.147$, $p = 0.067$). The rainy scenario (S3) yields the highest reported effort, suggesting increased perceived workload under low-visibility conditions. The baseline scenario (S1) does not show a distinct trend in subjective ratings.

4.3.4 Summary of RQ1 Findings. Across modalities, hazardous scenarios significantly influence drivers’ attention, longitudinal driving behavior, and perceived workload. The rainy scenario (S3) emerges as one of the most challenging conditions, followed by the nighttime (S5) and curved-road (S4) scenarios. In contrast, the baseline scenario (S1) consistently shows stable attention patterns, smoother driving behavior, and lower workload, supporting its role as the safest reference condition.

4.4 Work-Zone Design Effects

4.4.1 Attention Metrics. The ANOVA results indicate significant effects of work-zone design on gaze-based attention measures. As shown in Fig. 11(a), gaze duration varies across work-zone types. The work zone with an advance HUD warning displayed 100 meters ahead (WZ3) exhibits the longest gaze duration, suggesting that early visual warnings encourage drivers to attend to the work zone earlier and for a longer period of time.

Table 2. One-way ANOVA results for scenario-level (RQ1) and work-zone-level (RQ2) effects.

RQ1: Scenario Effects			RQ2: Work-Zone Design Effects		
Metric	F	<i>p</i>	Metric	F	<i>p</i>
Gaze duration	3.738	0.003**	Gaze duration	3.937	0.020**
Gaze fixation ratio (R50)	1.948	0.086*	Gaze fixation ratio (R50)	2.689	0.069*
Gaze fixation ratio (R100)	17.513	<0.001***	Gaze fixation ratio (R100)	33.004	<0.001***
Gaze fixation ratio (R120)	44.296	<0.001***	Gaze fixation ratio (R120)	47.346	<0.001***
Speed mean	7.893	<0.001***	Speed mean	1.598	0.204
Speed standard deviation	4.012	0.001**	Speed standard deviation	3.634	0.027**
Acceleration mean	13.010	<0.001***	Acceleration mean	4.403	0.013**
Acceleration standard deviation	3.122	0.009**	Acceleration standard deviation	1.843	0.160

* $p < 0.1$, ** $p < 0.01$, *** $p < 0.001$.

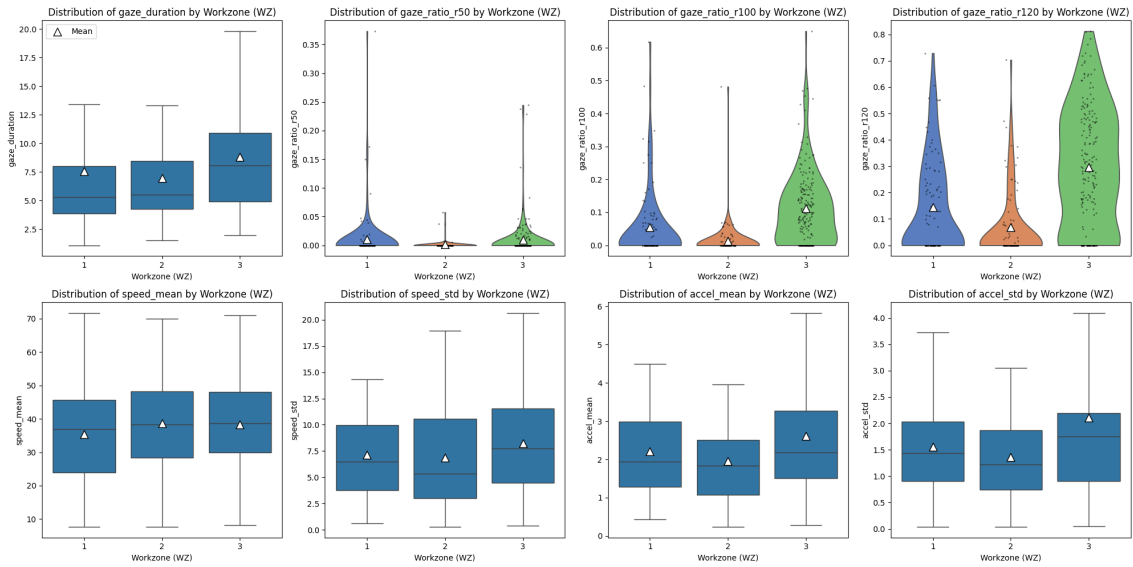


Fig. 11. Work-zone-level distributions of gaze-based attention and driving behavior across three work-zone designs (WZ1–WZ3). Triangles denote mean values.

Figures 11(b–d) show gaze fixation ratios computed at three spatial scales (R50, R100, and R120). At the smallest radius (R50), fixation ratios remain close to zero across all work-zone types, with only marginal differences. At larger radii (R100 and R120), fixation ratios differ more clearly across work zones. WZ3 consistently elicits higher fixation ratios than the other designs, indicating that advance warnings increase drivers’ visual attention toward the work-zone region.

Interestingly, the VR-worker work zone (WZ2) receives lower fixation ratios than the baseline work zone (WZ1), despite the presence of a dynamic and interactive worker. This result is unexpected, as human motion and interaction are typically assumed to attract greater visual attention. Instead of sustained fixation on a single work-zone element, the VR-worker may induce different visual strategies, such as more frequent gaze shifts among the worker, traffic

cones, and roadway boundaries. This surprising finding suggests that interactive work-zone elements do not necessarily translate into higher fixation-based attention and highlights the complexity of driver attention allocation in interactive environments.

4.4.2 Driving Behavior Metrics. Driving behavior metrics provide complementary insight into how work-zone design influences vehicle control. As shown in Fig. 11(e-f), mean speed does not differ significantly across work-zone types, indicating that participants maintained similar average speeds when passing different work zones. In contrast, speed standard deviation shows significant differences, reflecting variations in speed control strategies across work zones and participants.

Figures 11(g-h) present acceleration metrics across work-zone types. Acceleration mean shows a significant difference, with the VR-worker work zone (WZ2) exhibiting the lowest acceleration variability. Although the VR worker occasionally interacts with vehicles by placing traffic cones and altering the work-zone boundary, drivers tend to respond primarily through lateral steering adjustments rather than frequent braking or throttle inputs. As a result, WZ2 is associated with relatively smooth longitudinal control despite its interactive nature. Acceleration standard deviation does not show a significant difference across work-zone types.

4.4.3 Self-Reported Cognitive Load. After completing all driving tasks, participants filled out a post-experiment questionnaire evaluating different work-zone designs in terms of clarity, predictability, and attention required. No significant differences are observed across work-zone types for any self-reported metric. All work zones receive relatively high scores, indicating that participants generally perceived all designs as understandable and reasonable. Notably, even the presence of an advance HUD warning did not lead to clear differentiation in subjective ratings, suggesting that self-reported measures may be less sensitive to subtle differences in work-zone design than objective attention and driving behavior metrics.

4.4.4 Summary of RQ2 Findings. Overall, the results indicate that work-zone design influences drivers' visual attention and driving behavior, even when such differences are not reflected in self-reported evaluations. The advance HUD warning work zone (WZ3) consistently attracts greater visual attention, demonstrating the effectiveness of early warnings in guiding driver perception. In contrast, the VR-worker work zone (WZ2) produces a set of unexpected effects, including lower fixation ratios, higher mean speed, and smoother longitudinal control. Although participants do not explicitly distinguish among work-zone types in subjective reports, objective measures reveal clear differences in how drivers allocate attention and control their vehicles. These findings highlight the value of multimodal, measurement-based evaluation for understanding driver responses to interactive and emerging work-zone designs.

4.5 Users' Feedback on DUALSense

DUALSense is a user-centered platform designed to collect human driving behavior data for roadway work zone safety and autonomous driving research. Since the platform relies on drivers' natural reactions and behaviors, collecting user feedback is essential for evaluating its usability and overall driving experience.

In the post-study questionnaire, the first section focused on participants' perceived system experience. Three aspects of the driving simulator were evaluated on a 5-point Likert scale: *naturalness of driving interaction*, *stability and smoothness of the system*, and *motion sickness*. The average ratings were 3.25/5 for naturalness, 3.56/5 for stability and smoothness, and 1.25/5 for motion sickness. These results indicate that participants generally perceived the system as

natural and stable, while experiencing minimal simulator-induced discomfort, suggesting that DUALSense provides a reliable and user-friendly driving environment.

In addition to the quantitative ratings, participants provided qualitative feedback on their system experience. Most participants did not report major issues or suggest substantial changes to the platform. Several participants commented that vehicle motion felt natural after reaching driving speed, and that steering, throttle, and brake inputs were intuitive. However, a small number of participants suggested that vehicle handling and acceleration pedal behavior could be further refined.

Participants were also asked about their experiences across the six driving scenarios. Rainy conditions (S3) were most frequently mentioned as the most challenging scenario, followed by curved roads (S4) and nighttime driving (S5). These scenarios were reported to require increased attention due to reduced visibility or limited sight distance. Feedback regarding advance V2I warning signage in WZ3 was mixed: some participants appreciated additional warnings for improving perceived safety, while others reported that the warning signage could obstruct forward visibility.

When asked which elements were most helpful for driving, participants commonly mentioned steering control, gas and brake pedals, and speed notifications as important for vehicle control. Road signage was also frequently cited as helpful for recognizing the presence of work zones. In contrast, suggested improvements focused on traffic signal behavior (e.g., very short or missing yellow phases), the absence of brake light activation on leading vehicles, lack of left and right mirrors, and head-up display (HUD) warnings that occasionally blocked the driver’s view.

5 Conclusion and Discussion

5.1 Discussion

The user study reveals differentiated risk levels across hazardous roadway work zone scenarios. Among all scenarios, the rainy condition (S3) consistently emerges as the most challenging, as supported by attention-related metrics, driving behavior measures, and both qualitative and quantitative user feedback. Curve roads (S4) and nighttime driving (S5) also exhibit elevated risk levels; however, not all metrics consistently align to the same extent as observed in the rainy condition, suggesting more moderate or scenario-dependent effects.

Regarding work zone design, the head-up display (HUD) warning was found to increase drivers’ attentional engagement based on objective measurements. However, user feedback toward this design was polarized. While some participants appreciated the additional warnings and perceived increased safety, others reported that the HUD elements occasionally obstructed their forward visibility. These findings suggest that HUD-based warnings hold promise for improving work zone safety, but their visual design and presentation require further refinement to balance effectiveness and usability.

An interesting observation is that the VR-worker condition elicited comparatively lower attentional demand and smoother driving behavior. Compared to prerecorded animated workers, the VR worker exhibited a larger interaction space and more complex, less predictable movements. Despite this increased behavioral complexity, drivers appeared to adapt well to this work zone type. Future works need to cover the behavior analysis about WZ type 2, by considering worker’s gaze and trajectory with driver and vehicle side.

5.2 Conclusion

This paper introduces **DUALSense**, a VR-based dual-user interaction platform designed to safely collect multimodal data in a wide range of hazardous roadway work zone scenarios. The platform provides a flexible framework for

studying human interactions under safety-critical transportation conditions and can be extended to include additional vulnerable road users, offering a low-cost and safe alternative to real-world data collection.

Through a controlled usability test, we recreated five high-risk roadway work zone scenarios and examined drivers’ reactions and performance across scenarios. Results indicate that rainy conditions pose the highest perceived risk, followed by nighttime and curved-road scenarios, addressing RQ1 by demonstrating scenario-dependent differences in driver attention and perceived difficulty.

The study further investigates different work zone designs, addressing RQ2 by showing that design choices such as HUD-based warnings can increase drivers’ attentional engagement, although usability trade-offs related to visual interference remain. Overall, these findings highlight the value of human-centered evaluation in understanding how both scenario characteristics and work zone designs shape driver experience. Taken together, the results demonstrate the potential of DUALSense as a research platform for studying roadway safety and informing future human-centered autonomous driving systems.

5.3 Limitations and Future Work

Despite generally positive feedback from participants, several limitations remain. Some system-level aspects could be further refined, including pedal control responsiveness, traffic signal behavior (e.g., missing or short yellow phases), the absence of brake light activation on leading vehicles, limited side mirrors, and overall vehicle control smoothness. Although HUD warnings were effective in increasing attention, mixed user responses highlight the need for further investigation into visual design strategies that enhance user trust and acceptance.

Future work will involve larger-scale user studies to improve statistical power and generalizability. Given the rich multimodal data collected—including direct human signals (e.g., gaze and biometrics) and indirect vehicle-state data—there is a need for a unified framework to align and integrate these modalities. In particular, linking physiological responses to subjective feedback and driving behavior remains an open challenge. Additionally, defining meaningful thresholds for assessing risk levels across different hazardous scenarios warrants further exploration. Another promising direction is to investigate collaborative interactions between human drivers and autonomous vehicles using the proposed platform and dataset to further enhance roadway safety.

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This appendix provides methodological details to support reproducibility and transparency, complementing the main manuscript.

A Appendix A: VR Worker Development Steps

Appendix A provides visual illustrations of the VR-worker development steps through figures. Figure 12 shows (a) full-body movement calculation, (b) lower-body animations, and (c) grabbing with motion controllers, while Figure 13 presents (d) pre-recorded worker animations. These components are shown visually rather than described in text.

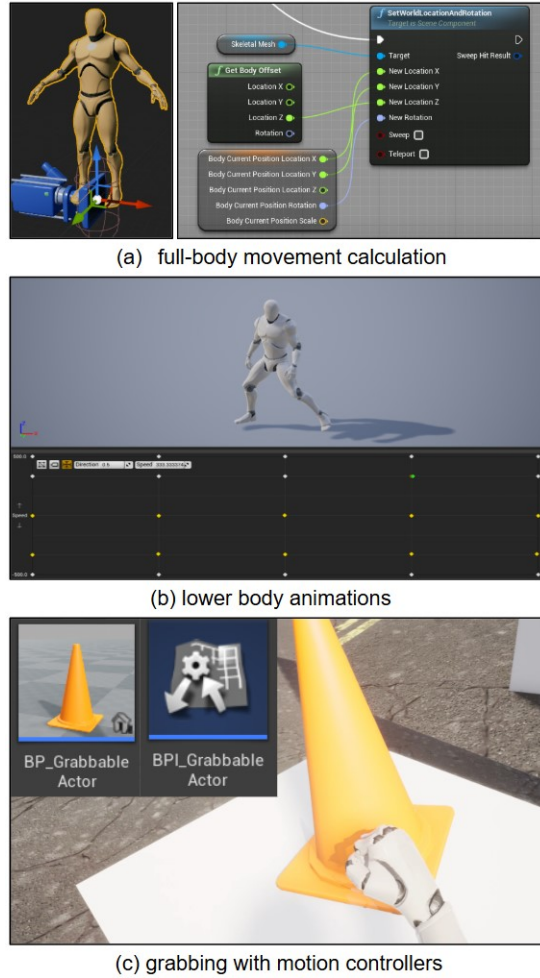


Fig. 12. VR-worker development modules: (a) full-body movement calculation, (b) lower body animations, (c) grabbing with motion controllers

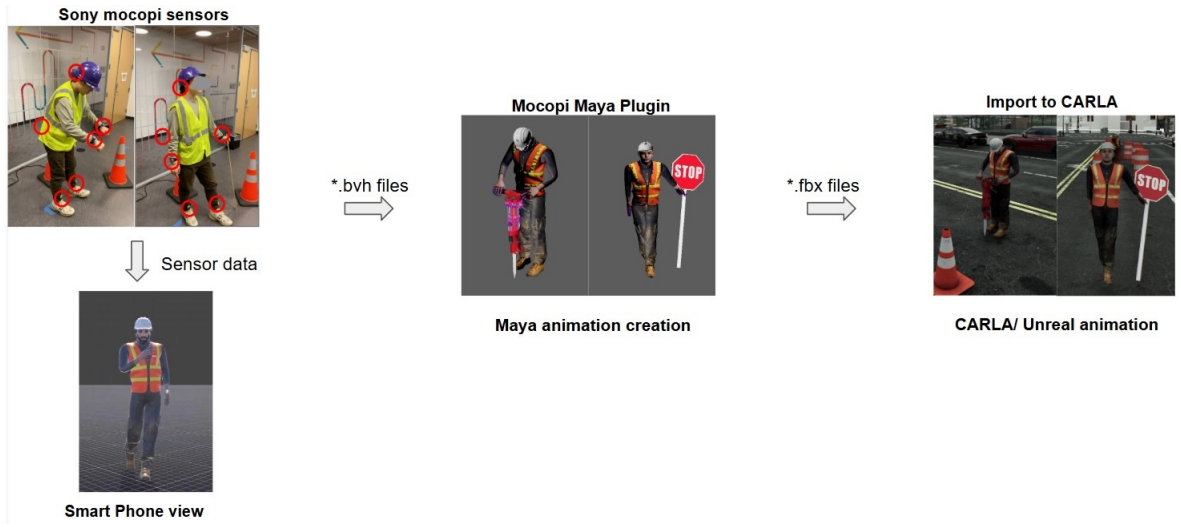


Fig. 13. VR-worker development modules: (d) pre-recorded various worker animations

B Appendix B: Sensor data collection & architecture

The architecture of VR-worker and drivers and vehicles is asyn mode to maximize the rendering ability and best performance for human users. The VR FPS is between 40 - 70 depending on scenarios complexity. And the driving ui is at least 35. The architecture is shown in Figure 14.

The other sensor includes LiDAR, trajectory, speed and acceleration and steering controls show visualization in Figure 15 and Figure 16

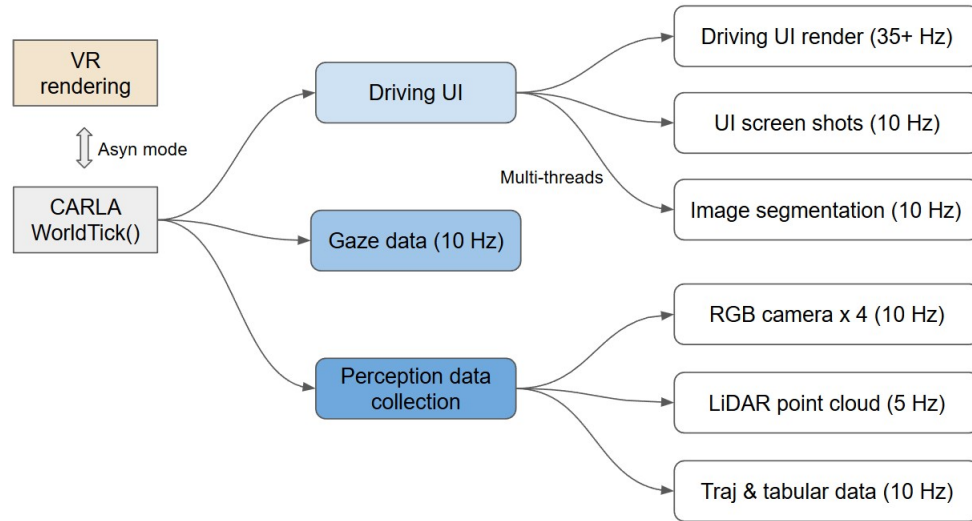


Fig. 14. Architecture of data collection and driving UI built on the CARLA PythonAPI

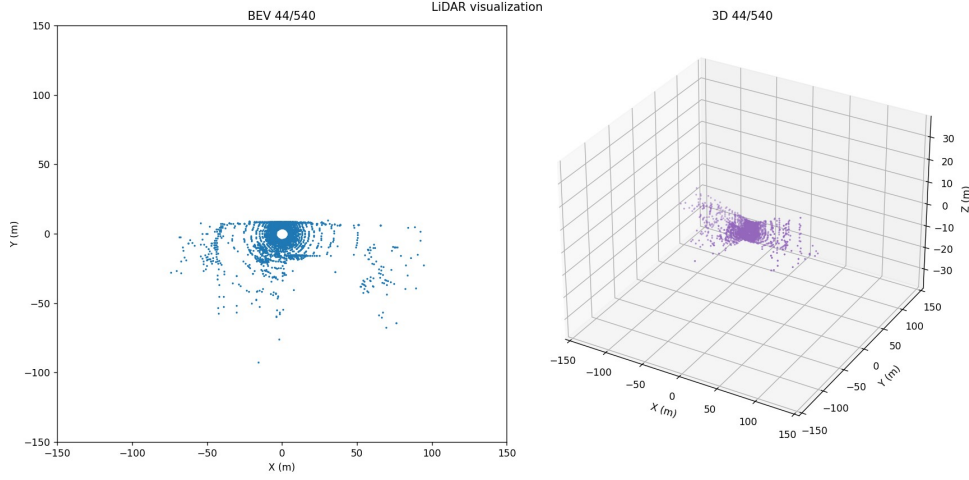


Fig. 15. Sensor data collection: LiDAR point clouds visualizations

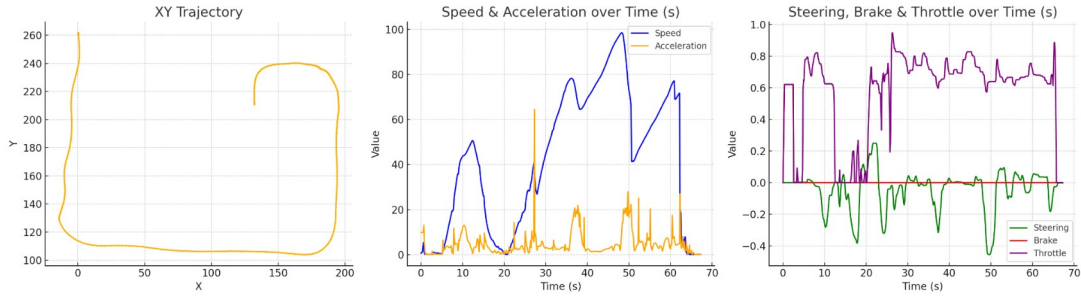


Fig. 16. Sensor data collection: Trajectory, speed and human control commands visualizations

C Appendix C: NASA-TLX and Post-Study Questionnaires

C.1 NASA-TLX and Task Load Assessment

To assess perceived workload during the driving tasks, we administered a NASA-TLX-based questionnaire using a 5-point Likert scale. The questionnaire included the following dimensions:

- *Mental Demand*: How mentally demanding was the driving task? (1 = very low, 5 = very high)
- *Physical Demand*: How physically demanding was the driving task? (1 = very low, 5 = very high)
- *Temporal Demand*: How hurried or rushed did the pace of the task feel? (1 = not at all, 5 = very rushed)
- *Performance*: How successful do you think you were in completing the task? (1 = very poor, 5 = very good)
- *Effort*: How hard did you have to work to achieve your performance level? (1 = very easy, 5 = very hard)
- *Frustration*: How insecure, irritated, stressed, or annoyed did you feel during the task? (1 = not frustrated at all, 5 = very frustrated)

In addition to the standard NASA-TLX dimensions, we included two task-specific measures to capture driving-related workload in roadway work zone scenarios:

- *Scene Clarity*: How clear and understandable were the road elements (e.g., lane markings, signs, cones, vehicles)? (1 = very unclear, 5 = very clear)
- *Attention Requirement*: How much attention was required to complete the driving task smoothly? (1 = very little, 5 = very high)

C.2 Post-Study Questionnaire (Excerpt)

A. System Experience

- *Overall ease of use*: How easy or difficult was the driving system to use? (1 = very easy, 5 = very hard)
- *Naturalness*: How natural did the driving interaction feel (steering, pedals, view control)? (1 = very artificial, 5 = very realistic)
- *Stability and smoothness*: How stable and smooth did the system feel during the session? (1 = very unstable, 5 = very smooth)
- *Motion sickness*: To what extent did you experience motion sickness? (1 = not at all, 5 = very much)

B. Differences Across Driving Scenarios

Participants were asked to reflect on perceived differences across six scenarios (Normal, Nighttime, Rainy, Curve, Truck, No Signage):

- Did you experience differences in driving difficulty across scenarios? If so, which scenarios felt more difficult and why?
- Did the clarity of road elements (e.g., lanes, signage, vehicles, pedestrians) differ across scenarios?
- Were there scenarios that required greater concentration than usual?

C. Comparison of Work Zone Designs

Three work zone designs were evaluated: WZ-A (normal), WZ-B (VR worker), and WZ-C (advance warning at 100 m).

- *Clarity*: How clear was each design? (1 = not clear, 5 = very clear)
- *Predictability*: How predictable was each design? (1 = not predictable, 5 = very predictable)
- *Attention required*: How much attention did each design require? (1 = little attention, 5 = very high attention)

D. Open-Ended Questions

- Which system features or design elements felt most helpful or natural during the experiment?
- Were there any parts of the system or scenarios that confused you or could be improved?

E. Background Information

- Self-reported driving experience (years and/or mileage).

C.3 Post-Study Subjective Evaluations

In addition to task-level workload measures, we report post-study subjective evaluations to characterize participants' overall experience and perceptions of the simulated driving scenarios. These measures include self-reported cognitive load dimensions derived from NASA-TLX, perceived work-zone clarity, predictability, and attention demand across different work-zone designs, as well as overall system experience metrics such as perceived naturalness of interaction, system stability and smoothness, and motion sickness. All responses were collected using 5-point Likert scales, and the distributions are presented to provide descriptive insights into participants' subjective experiences.

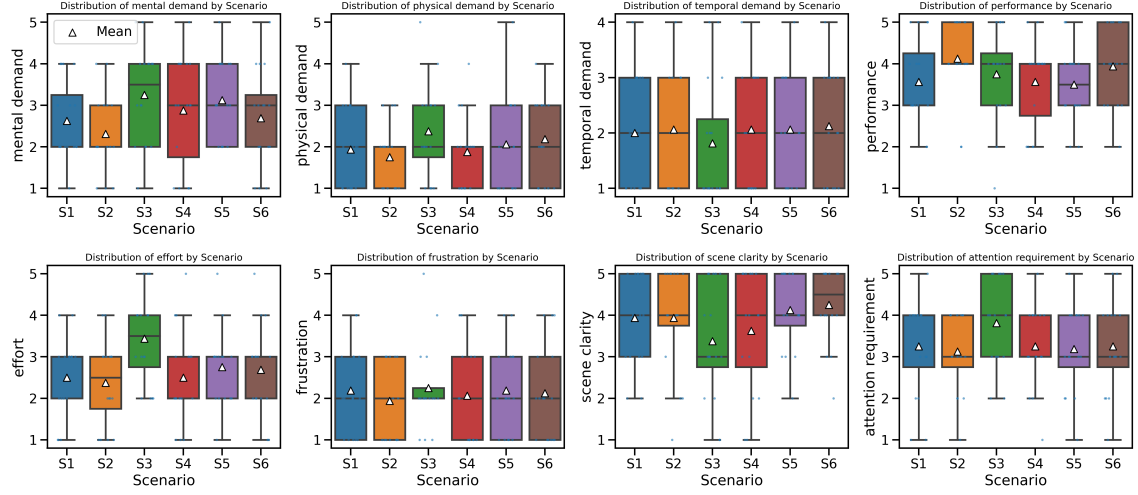


Fig. 17. Scenario-level distributions of self-reported cognitive load measured by NASA-TLX. Subplots (a)–(h) show ratings of mental demand, physical demand, temporal demand, performance, effort, frustration, scene clarity, and attention requirement across six driving scenarios (S1–S6). Triangles indicate scenario-level means.

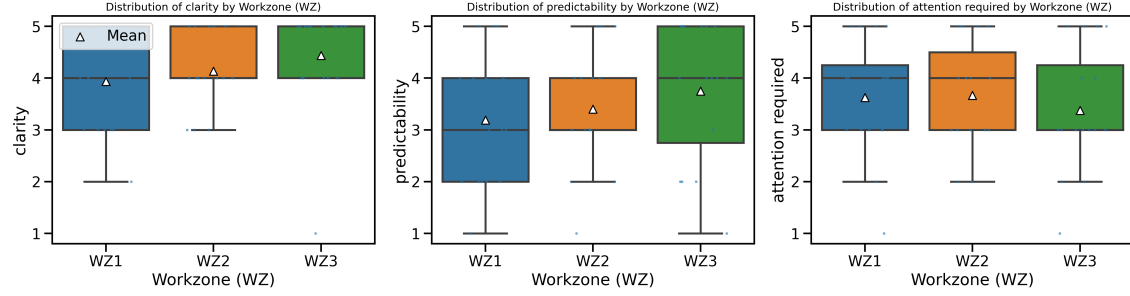


Fig. 18. Post-study self-reported evaluations of work-zone designs (WZ1–WZ3), including clarity, predictability, and attention required.

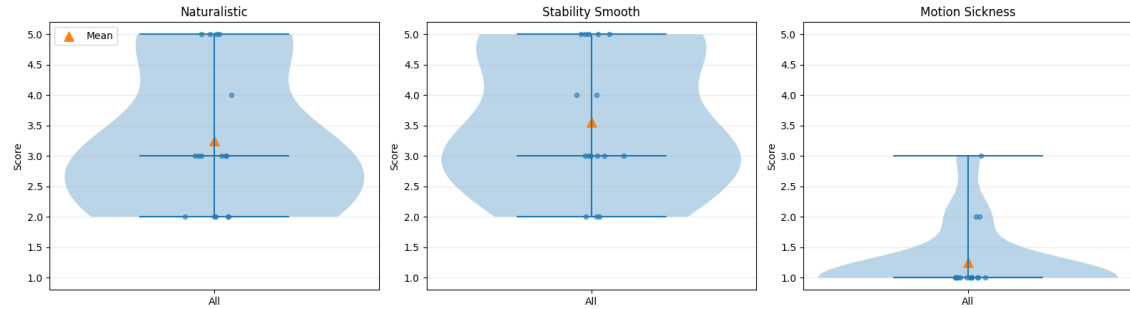


Fig. 19. Post-study self-reported system experience, including perceived naturalness of driving interaction, system stability and smoothness, and motion sickness.